

Grade 7
Unit 11
Lesson 9 Practice

Name Answer Key
Date _____
Period _____

1. Complete the table to find the area and circumference of a circle when given its diameter. $C = d\pi$ or $2\pi r$ $A = r^2 \cdot \pi$

Diameter	3	9	13	18	22
Circumference	3π	9π	13π	18π	22π
Area	2.25π	20.25π	42.25π	81π	121π

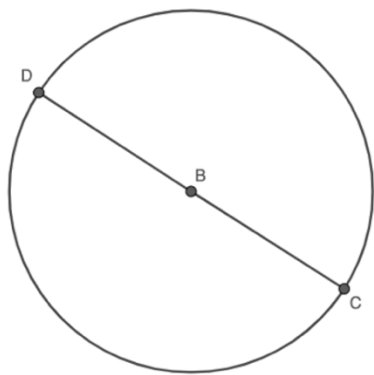
2. Complete the table to find the area and circumference of a circle when given its radius.

Radius	2	11	23	33	36
Circumference	4π	22π	46π	66π	72π
Area	4π	121π	529π	$1,089\pi$	$1,296\pi$

Complete the statements.

3. There ☒ is ☐ is not a proportional relationship between the diameter and circumference of a circle,
- ☐ Therefore, there is no constant of proportionality.
☒ The constant of proportionality is π .
4. There ☒ is ☐ is not a proportional relationship between the radius and circumference of a circle.
- ☐ Therefore, there is no constant of proportionality.
☒ The constant of proportionality is 2π .
5. There ☐ is ☐ is not a proportional relationship between the radius and area of a circle.
- ☒ Therefore, there is no constant of proportionality.
☐ The constant of proportionality is _____.

Circle B is shown with diameter of CD .



6. Tomas uses a drawing program to change the size of Circle B . Find the circumference, in terms of π , for each diameter he uses.

Diameter	Circumference
14	14π
25	25π
47	47π
137	137π

7. Suppose x represents the diameter and y represents the circumference. Determine the constant of proportionality, $\frac{y}{x}$.

$$\frac{14\pi}{14} = \pi$$

8. Complete the statement.

As the value of the diameter increases by 2, the value of circumference increases by 2π .

9. Write an equation that represents the relationship between x , the diameter, and y , circumference.

$$y = \pi x$$